

on a 30/360 basis. Referring to Equations 4.8 and 4.11, the underlying is the 1-year LIBOR forward that sets at time t_i , and we have $\Delta t = 1$ year = 360 days and $\tau = 1$. The value of a caplet that resets at time t_i and pays off at time t_{i+1} is

$$\frac{1}{[1 + \text{zc}(t_{i+1})]^{t_{i+1}}} \times [f(t_i, t_{i+1})N(d_1) - XN(d_2)], \quad (4.12)$$

where

$$d_1 = \frac{\ln(f(t_i, t_{i+1})/X) + \sigma_{t_i}^2 \times t_i/2}{\sigma_{t_i} \times \sqrt{t_i}} \quad (4.13)$$

and

$$d_2 = d_1 - \sigma_{t_i} \times \sqrt{t_i}. \quad (4.14)$$

The value of a floorlet that resets at time t_i and pays off at time t_{i+1} is

$$\frac{1}{[1 + \text{zc}(t_{i+1})]^{t_{i+1}}} \times [XN(-d_2) - f(t_i, t_{i+1})N(-d_1)], \quad (4.15)$$

with d_1 and d_2 as given in Equations 4.13 and 4.14, respectively. The quantities

$$[f(t_i, t_{i+1})N(d_1) - XN(d_2)]$$

and

$$[XN(-d_2) - f(t_i, t_{i+1})N(-d_1)]$$

are often referred to as the *forward premium* for the caplet and the floorlet, respectively. To determine the premium in dollars for caplets and floorlets, we multiply Equations 4.12 and 4.15, respectively, by the notional.

Example 1

Let's take a look now at pricing up a cap in our stylized model. Consider a \$100mm 4-year cap on 1-year LIBOR struck at 5.47%. Assume that the swap curve is as in Table 4.2 and further assume the base vol curve is flat at 15%.

Figure 4.4 shows how we price up the cap. The first thing to note is that we do not use Black's formula to value the spot starting caplet. There is no option associated with this caplet; the LIBOR reset has already occurred. Since the LIBOR set of 5% is below the strike, the value of the first caplet is zero. For each of the remaining caplets, we first compute d_1 and d_2 as per Equations 4.13 and 4.14, respectively, and then we compute the forward premium. We use the appropriate zero coupon rate to compute the value of the caplet, then multiply by the notional to get the value of the caplet in

Maturity	Treasury Yield	Swap Spread	Swap Rate (Ann 30/360)
1	4.80%	20	5.00%
2	4.88%	23	5.11%
3	5.05%	25	5.30%
4	5.19%	28	5.47%

Table 4.2: The Swap Curve

Cap model									
							Forward		Caplet
Time	zc	Period	Forward	σ	d_1	d_2	premium	Caplet	in dollars
1	5.000%	0x1	5.000%	n/a	n/a	n/a	0.000%	0.000%	0
2	5.113%	1x2	5.226%	15%	-0.230	-0.380	0.212%	0.192%	192,131
3	5.312%	2x3	5.712%	15%	0.310	0.098	0.603%	0.516%	516,411
4	5.494%	3x4	6.041%	15%	0.512	0.252	0.923%	0.745%	745,442
Strike 5.470%		Notional			100,000,000			1,453,985	
								in bps	145.4

Figure 4.4: Pricing Up a 4-Year Cap on 1-Year LIBOR with a Strike of 5.47%

dollars. The sum of the caplet values gives us the value of the cap. The value of the cap is 1.454%, or 145.4 bps of the notional. On a notional of \$100mm, the premium in dollars is \$1,453,984.¹⁷ □

Example 2

The value of a \$100mm 4-year floor on 1-year LIBOR struck at 5.47% is found very similarly as shown in Figure 4.5. The only thing to note here is that the initial LIBOR setting of 5% is in-the-money. We calculate the first

¹⁷When referring to the prices of caps and floors, it is most common to quote prices in bps of the notional. Prices are often rounded to the tenth of a bp, or to quarter bp increments. Thus, the price quoted for an option in bps most often does not exactly match the dollar price of an option from a model. The rounding error is typically small and safely ignored in industry. In the example we have considered here, the premium would likely be quoted as 145.4 bps, or \$1,454,000 in dollars.

Floor model									
							Forward		Floorlet
Time	zc	Period	Forward	σ	d_1	d_2	premium	Floorlet	in dollars
1	5.000%	0x1	5.000%	n/a	n/a	n/a	0.470%	0.448%	447,619
2	5.113%	1x2	5.226%	15%	-0.230	-0.380	0.457%	0.413%	413,199
3	5.312%	2x3	5.712%	15%	0.310	0.098	0.361%	0.309%	309,043
4	5.494%	3x4	6.041%	15%	0.512	0.252	0.352%	0.284%	284,123
Strike	5.470%		Notional			100,000,000			1,453,985
								in bps	145.4

Figure 4.5: Pricing Up a 4-Year Floor on 1-Year LIBOR with a Strike of 5.47%

floorlet's value by hand. Since the strike is 5.47% and the LIBOR setting for the first period is 5%, the first floorlet will pay out $5.47\% - 5.00\% = 47$ bps. This amount will be received in one year. To compute the present value of this amount simply multiply it by the discount factor formed from the 1-year zero coupon rate

$$\frac{1}{[1 + zc(1)]^1} \times 0.47\% = \frac{1}{1.05} \times 0.47\% = 0.448\%$$

and then multiply by the notional to get the value of this initial floorlet in dollars: $\$100\text{mm} \times 0.00448 = \$447,619$.

Notice that the value of the 4-year cap with a strike of 5.47% is the same as the value of 4-year floor with a strike of 5.47%. This is not a coincidence. From cap-floor parity, we know that the value of a cap less the value of a floor with the same strike must be equal to the value of a swap whose fixed rate is equal to the strike. But since 5.47% is the 4-year swap rate, this means that a swap with a fixed rate of 5.47% is worth zero. Therefore, a 4-year cap and a 4-year floor with a strike equal to the 4-year swap rate of 5.47% must be worth the same. $\square$

4.1.7 Variations of Standard Caps and Floors

The market has developed numerous variations of standard caps and floors that we have focused on thus far. Sometimes these have been created so as to provide a lower cost alternative to a standard option. Other times these

have been created in order to better fit the view that an investor has and wants to take, or to better hedge a risk that he faces.

The first two trades that we will look at in this section, step-up strike and periodic caps, are somewhat common variants. The latter two are extremely rare. The purpose of discussing them is to demonstrate the flexibility that the OTC derivative markets have afforded market participants.

Step-Up Strike Caps

As the name suggests, *step-up* strike caps are caps in which the strike is not the same for each and every caplet. On the trade date, the counterparties specify the strike for each of the caplets in the trade. For example, in our stylized example, one might specify the strikes in a 4-year cap are 5%, 5.25%, 5.50%, and 5.75%. This is priced using a slight modification to Black's formula in which the appropriate strike is used for each caplet. Step-up caps are often used to decrease the cost of a cap.¹⁸ In exchange for the lower cost, the buyer is giving up some protection the cap may provide if it had the lower initial strike over its life. Floors with strikes that vary each floorlet have also traded.

Periodic (Sticky) Caps and Floors

The easiest way to demonstrate the nature of *periodic caps or floors* is to discuss an asset that has a periodic option embedded in it. Consider a 5-year bond that pays annual 30/360 coupons in accordance with the following formula:¹⁹

Year 1:	6%
Years 2–5:	1-year CMT + 100 bps, subject to the condition that the coupon can be no more than 200 bps greater than the coupon paid in the prior year.

Assume that 1-year CMT, the 1-year Constant Maturity Treasury, is reset at the beginning of each of the last four years for purposes of computing the coupon paid in years two through five, respectively.

A series of options have been embedded that constrain the coupons that can be paid by the bond. For example, suppose that in one year 1-year CMT

¹⁸Step-up strike caps have typically been popular in a steep swap curve environment in which forward LIBOR rates are relatively high, making the cost of caps more expensive than in a flatter curve environment, *ceteris paribus*.

¹⁹Those who are familiar with mortgage securities will note that this fictitious bond is reminiscent of an ARM, or adjustable rate mortgage.

sets at 20%. The highest coupon that the bond can pay in year two is 8%, not the fully indexed amount of 1-year CMT + 100 bps = 20% + 1% = 21%. These embedded options therefore constrain the coupons, which hurts the bondholder. That is, the bondholder is short these options. Note that these options constitute some sort of cap: they are in-the-money when 1-year CMT goes above some level. Thus, the embedded options are a cap on 1-year CMT.

Now let's look at the strikes on each of the caplets in this trade. Standing here today, can we say for certain what the strike on each caplet is going to be? The answer is: it depends. It depends on what year of the trade we are talking about. For example, in year two, which has the first embedded caplet, we can definitively determine what the strike on the caplet is. We know that in year two the cap will kick in whenever 1-year CMT + 100 bps > 6% + 2%, that is, whenever 1-year CMT is greater than 7%. But what about in the subsequent years? Can we definitively say today what the strike on the remaining caplets will be? The answer is no, because in each of the subsequent years the answer depends on where 1-year CMT set previously.

For example, let's look at the coupon paid in year three, which is of course equal to 1-year CMT + 100 bps, subject to the condition that the coupon can be no more than 200 bps greater than the coupon paid in the prior year. Suppose that at the beginning of year two, 1-year CMT was 2%. Then the coupon in year two was 3% according to the formula. Now let's look at the highest coupon that can be paid in the third year. Since the coupon paid in year two is 3%, the highest coupon that can be paid in year three is 5%. Thus, the caplet in year three kicks in whenever 1-year CMT + 100 bps > 3% + 2%, or when 1-year CMT is greater than 4%. Thus the strike on the caplet in year three is 4% in this case.

But suppose now that at the beginning of year two 1-year CMT was 10%. Then the coupon paid in year two was 8% (which is equal to 1-year CMT + 100 bps, subject to the condition that the coupon is no greater than 200 bps more than the coupon paid in the prior year). Now let's look at the highest coupon that can be paid in the third year. Since the coupon paid in year two is 8%, the highest coupon that can be paid in year three is 10%. Thus, the caplet in year three kicks in whenever 1-year CMT + 100 bps > 8% + 2%, or when 1-year CMT is greater than 9%. Thus the strike on the caplet in year three is 9% in this case. These findings are summarized in Table 4.3.

Thus we cannot say definitively today what the strikes on the caplets are. It depends on the path of 1-year CMT over the life of the trade. These types of caps are known as *sticky* or *periodic* caps. The word "sticky" derives from the nature of the coupons paid by this bond: they can go up, but only so fast. If 1-year CMT were to go up dramatically each year, the coupons

1-Year CMT in Year 2	Coupon in Year 2	Highest Possible Coupon in Year 3	Strike on Embedded Caplet in Year 3
2%	3%	5%	4%
10%	8%	10%	9%

Table 4.3: Strike on Embedded Caplet in Year 3 of Asset as a Function of 1-Year CMT Setting in Year 2

would go up but would be sticky from one year to the next, only able to go up so much in any one year. Likewise, the embedded option has a strike that is sticky as well.

Sticky caps and floors are exotic options whose pricing is beyond the scope of this discussion. Their usage is somewhat limited these days, but they have been very popular at times. Given their relatively complex nature, it was not unusual to observe very disparate prices for these options among dealers when they first became popular in the 1990s.

Chooser Caps

A *chooser cap* is a cap in which the holder has the right to exercise a given number of caplets over the life of the cap. On each reset date, the holder will observe where 3-month LIBOR sets and will decide whether or not to exercise the caplet, assuming that he has not already exercised the maximum number of caplets allowable under the terms of the contract.²⁰

As a specific example, consider a 5-year 50% chooser cap on 3-month LIBOR struck at 6%. The holder will have the right to exercise any 10 of the 20 caplets that make up the regular cap. The holder does not choose today but waits until the reset on each caplet and, based on the reset, decides whether or not he wants to select a payment from that caplet. Once the holder has selected payment from any 10 caplets, the trade is over, regardless of the stated 5-year final maturity.

The motivation behind such a trade is as follows: in the course of normal business cycles, rates will move up and down over time. As such, the pur-

²⁰This is not an industry standard name. I would suggest that you not walk around dropping the name “chooser cap” and expect that people will know what you are talking about. In fact, I am only aware of a chooser cap trading once. Despite the infrequency with which this product trades, it is a good example demonstrating the flexibility that exists in structuring OTC derivative trades.

chaser of a cap may expect that some caplets will finish in-the-money, but some will finish out-of-the-money. A chooser cap will be a cheaper alternative to a standard cap of the same maturity, providing selective protection for certain periods of time over the life of the standard cap. The holder of a 5-year chooser cap will still have protection against higher rates until the same final maturity as a regular 5-year cap offers but will choose the exact periods of time during which the protection will be invoked. If the cap buyer believes that some of the caplets will finish out-of-the-money anyway, why should he bother to pay for them all? The risk associated with buying a chooser cap instead of a standard cap is that rates move above the strike and stay above the strike, in which case the limited number of choices associated with a chooser cap does not provide as much protection as a regular cap. In this scenario, the buyer of a chooser cap will have paid a smaller upfront premium but would not be able to get payments from each of the in-the-money caplets.

Note that the holder of a chooser cap would never select a payment from a caplet whose reset is out-of-the-money. He would forgo choosing such a caplet with zero payoff in order to maintain a choice in a subsequent quarter, leaving the possibility for a positive payout. But even if a particular caplet is slightly in-the-money, it is not clear that the holder would elect to choose it. For example, consider again our example of a 5-year 50% chooser struck at 6%; suppose that on the first reset date that a caplet is in-the-money 3-month LIBOR has set at 6.01%. Since the payoff from this caplet is so small, perhaps the holder would rationally elect to forgo it, maintaining 10 choices for the future, rather than a small payoff in this current period and only 9 choices remaining.

Pricing up chooser caps is pretty complicated. We won't be able to give the pricing full justice here, but let's make a few observations. First, a chooser cap must be worth less than a regular cap of the same final maturity with the same strike. The payout each quarter from the regular cap will always be at least as much as the chooser cap—and may be more.

Now suppose that on the trade date we select any 10 of the 20 caplets in the regular cap and compare the total price of these 10 caplets to the chooser cap. The chooser cap must be worth at least as much as these 10 caplets chosen at random. If the chooser cap were worth less than the 10 caplets chosen at random, we could buy the chooser today and sell the 10 randomly selected caplets, taking in money today. Then we could use the chooser to select the exact 10 caplets that were selected at random, thereby perfectly covering our short, which locks in zero cash flows in the future. This would be an arbitrage, and thus the chooser must be worth at least as much.

We can actually come up with a more meaningful lower bound for the price of the chooser. Suppose now that we compare the value of the chooser

cap to the 10 most expensive caplets when we price up the cap today.²¹ The chooser cap must be worth at least as much as these 10 caplets even though they are the most expensive. If the chooser cap were less, then we would again buy the chooser and sell the 10 most expensive (which creates positive cash flow today), and we would use the chooser to cover our short perfectly, resulting in zero cash flows in the future, thus giving an arbitrage.²² In other words, the argument that we just constructed comparing the value of the chooser to any 10 caplets applies equally as well when those 10 caplets happen to be the most expensive ones in the cap.

We therefore have the following relationship between the price of a 50% 5-year chooser cap and the regular cap:

$$10 \text{ most expensive caplets} \leq 50\% \text{ chooser cap} \leq \text{regular cap.}$$

What's the difference between the 10 most expensive caplets and the chooser? We can think of the chooser as enabling the owner to get the payoff from the 10 most expensive caplets but also giving him the right to switch out of any or all of those 10 caplets, if over the course of time it becomes advantageous to do so. Let's generically refer to that option as a "switch option." We could then write the very loose relationship

$$50\% \text{ chooser cap} = 10 \text{ most expensive caplets} + \text{"switch option."}$$

As an example as to why this switch option may prove valuable, suppose that at the outset of the trade the 10 most expensive caplets are the last 10. However, assume that shortly after buying the chooser, the curve becomes extremely inverted. It could be that the back 10 caplets are worth very little, and the holder of the chooser may find it advantageous to take a payoff from any in-the-money caplets that come along.²³

²¹By the way, which caplets in a 5-year cap will be the 10 most expensive? Typically, they will be the last 10: in an upward-sloping curve environment, the first five years of the forward curve will often be upward sloping, and the combination of higher longer-dated forward rates and longer times to expiry usually means that the longer-dated caplets are worth more. It is, however, possible to construct examples in which the 10 most expensive caplets are not the last 10. If, for instance, the curve is inverted and/or the base vol curve is significantly inverted (meaning vols for longer-dated caplets are significantly below vols for shorter-dated caplets), it may be possible that longer-dated caplets are worth less than shorter-dated ones.

²²In fact, we could do better with the chooser: if any of the 10 most expensive caplets should expire out-of-the-money, we could forgo this choice and reserve it for one of the other 10 caplets instead if one should finish in-the-money, thereby performing better than the 10 most expensive caplets.

²³As indicated, the actual pricing of a chooser cap is quite complicated. Even the decision to choose a particular caplet for exercise is somewhat of an involved decision.

Convertible Cap Trade

Let's take a look now at the following trade, which, for lack of a better name, we will refer to as a *convertible cap* trade.²⁴ Consider a trade between a dealer and a client in which the dealer sells a 5-year cap on 3-month LIBOR struck at 7% to the client for some upfront premium. This cap is like a regular cap, except that in one year from today, the dealer has the right to

1. Call the cap from the counterparty *and*
2. Enter into a 4-year swap with the client in which the dealer receives 6% fixed and pays 3-month LIBOR.

Note that if the dealer calls the cap, then it must also enter into the swap. If the dealer does not call the cap, then it can't enter into the swap. The first question that we might ask is how does the premium for the convertible cap compare to the premium on a regular 5-year cap on 3-month LIBOR struck at 7%? The price of this convertible cap must be less, since the counterparty is buying the cap, but is simultaneously selling an option (i.e., to call the cap and enter into the swap) to the dealer.

The next question we might ask is, In what type of interest rate environment will the dealer want to exercise its option and call the cap and simultaneously enter into the swap? In order to address this question, let's first suppose that in a year rates are really, really high, say, 20% for all swap maturities. Then the cap will be worth a lot of money, so the dealer will be inclined to call the cap away from the counterparty. But not so fast. If rates are really high, then the dealer will not be so happy entering into a swap in which it is receiving only 6%. So it is not necessarily so clear whether or not the dealer will want to exercise its option in a high rate environment.

Now suppose that in a year rates are really, really low, like 2% across the curve. Well, in this case it seems like a good idea for the dealer to exercise its option in which case it will be in the swap receiving a high fixed rate of 6%. But, again, not so fast. If the dealer exercises its option, it will also take

Suppose that at a given point in time the holder of a chooser option has n remaining caplets that he can choose. On the next LIBOR reset date in the chooser cap, the holder of the chooser cap will need to compare i) the payoff from the current caplet plus the value of maintaining another $n - 1$ choices to be deployed among the remaining caplets and ii) forgoing a payment on the current caplet and maintaining n choices to be deployed among the remaining caplets. The decision to choose the current caplet will be made if the first of these alternatives is worth more.

²⁴This is also not an industry standard name by any means, nor is this trade at all standard. In fact, I have only seen it trade once. It was on *reverse inquiry*, meaning the terms of the trade were proposed by the client to the dealer, instead of by the dealer to the client.

delivery of the cap. But in a low interest rate environment, the cap is not going to be worth a whole lot. Usually when we think about exercising a call option and taking delivery of an underlying asset, we usually do so when the underlying is worth a lot, not a little. So it's not clear whether or not the dealer will want to exercise its option in a low rate environment.

So what to do? It's a little tough to figure out which way is up and which way is down in this problem. That is, it's not so clear in what type of rate environment the dealer is going to want to exercise its option. Let's break this problem down a bit more. From the point of view of the dealer,

Convertible Cap = Short 5-year cap on 3-month LIBOR at 7%
 + Long option that expires in one year to get a
 4-year 7% cap and simultaneously enter into a
 6% receive fixed 4-year swap.

Let's focus now on this option that the dealer is long. The dealer has the right, but not the obligation, to exercise the option and take delivery of a 4-year 7% cap; and if it does, it will have to enter into the 4-year swap in which it is receiving fixed at 6%. So this looks like a call option: the dealer can exercise and will take delivery of the underlying cap and will "pay a strike" that is equal to the swap it must enter into. Regarding this then as a call option on the cap with a strike equal to the swap, the payoff of this option at maturity (using slightly abusive notation) is equal to

$$\begin{aligned}
 \text{Option Payoff at Maturity} &= \max[\text{Cap} - \text{Strike}, 0] \\
 &= \max[\max(L - 7\%, 0) - \text{Strike}, 0] \\
 &= \max[\max(L - 7\%, 0) - (L - 6\%), 0] \\
 &= \max[\max(L - 7\%, 0) - (L - 6\% - 1\% + 1\%), 0] \\
 &= \max[\max(L - 7\%, 0) - (L - 7\%) - 1\%, 0] \\
 &= \max[\max(7\% - L, 0) - 1\%, 0] \\
 &= \max[\text{Floor struck at } 7\% - \text{annuity of } 1\%, 0].
 \end{aligned}$$

That is, the payoff of this option at maturity is equal to the maximum of zero or the value of a floor struck at 7% less the value of a 1% annuity running over the four years. Notice that we used a trick of both adding and subtracting 1% in the fourth line above, then invoked cap-floor parity in the sixth line. In this configuration, it is at least easy to determine in what type of interest rate environment the option will be exercised. The